

Advanced SQL Null functions.

Exercise 2.1.

Table 1: Employees

Syntax

SELECT employee-id, name,

COALESCE (Salary, 0) AS Salary-with-default

FROM Employees.

employee-id	name	Salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana.	0

Table 2: Orders. Question 2.

Syntax

SELECT employee-id, name,

COALESCE (department, 'Not Assigned')

AS department-name

FROM Employees;

employee-id	Name	department-name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana.	finance

Table 2: Orders

Syntax

SELECT order-id, Customer-id

FROM orders

Where Customer-id IS Null;

Order-id

Customer-id

103

Null.

Question 4

Syntax

```
SELECT order-id, Customer-id,  
COALESCE (delivery-date, 'Pending')  
AS delivery-status  
FROM Orders;
```

Order-id	Customer-id	delivery-status
101	201	2024-12-01
102	202	Pending
103	Null	2024-12-03

Table 3: Students

Syntax

```
SELECT Student-id, name,  
COALESCE (grade, 0) AS final-grade  
FROM Student;
```

Student-id	name	final-grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Question 6

Syntax

```
SELECT COUNT (*) AS not-graded-Count  
FROM Student  
WHERE grade IS Null;
```

not-graded-Count

1

Table 4: Products

Syntax

```
SELECT product-id, name, (Price - COALESCE  
(discount, 0)) AS final-price  
FROM Products;
```

Product-id	name	final-Price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Question 8 Table 5 Customers

Syntax

```
SELECT COUNT(*) AS Missing-email-Count  
FROM Customers  
WHERE email IS NULL;
```

Missing-email-Count
2

Question 9

Syntax

```
SELECT Customer-id, name,  
COALESCE(email, 'No Email') AS email-display  
FROM Customers;
```

Customer-id	Name	email-display
1	Linda	
2	Joseph	
3	Nia.	

Table 6 : Payment

Syntax

SELECT payment-id, COALESCE (Method, 'unknown')
AS Method-display, Status
FROM Payments;

Payment-id	Method-display	Status
301	Credit	Null
302	Paypal	Success
303.	unknown	failed

Table 7 inventory

Syntax

SELECT item-id, item-name,
COALESCE (quantity, 0) AS quantity-Checked
FROM Inventory ;

Item-id	item-name	quantity- Checked
1	Pen	0.
2	Notebook	150
3	Eraser	0

Table 8 Employees- Extra

Syntax.

SELECT emp-id, COALESCE (bonus, Commision)
AS first- available-reward
FROM Employees- Extra ;

emp-id	first-availale-reward
1	300
2	100
3	Null.

Table 9 Classes

Syntax

```
SELECT COUNT (*) AS no-room-Count  
FROM Classes  
WHERE room IS Null;
```

no-room-Count
2.

Table 10 Attendance

Syntax

```
SELECT Student-id, date,  
COALESCE (Status, 'Not Marked')  
AS attendance-Status  
FROM Attendance;
```

Student-id	date	attendance-Status
1	2023-04-01	Not Marked.
2	2023-04-01	Present
3	2023-04-01	Absent

Table 11 Bank-Accounts

Syntax

```
SELECT account-id,  
COALESCE (account-type, 'unknown') AS type-display  
COALESCE (balance, 0) AS balance-Checked  
FROM Bank-Accounts;
```

account-id	type-display	balance-Checked
A1	Saving	0
A2	Current	5000
A3.	Unknown	2000

Table 12 Projects

Syntax

SELECT Project-id, title,

COALESCE (Start-date, 'TBD') AS

Start-display

FROM Projects;

Project-id	title	
1	Website Revamp	2028-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13 Reviews

Syntax

SELECT review-id, product-id,

COALESCE (comment, 'No Comment')

AS comment-display, COALESCE (rating, 0)

AS rating display

FROM Reviews;

review-id	Product-id	Comment-display	rating-display
1	S01	Great product	4
2	S02	No Comment	0
3	S03	Works fine	3.

Table 14 Suppliers

Syntax

SELECT Supplier_id, name,
COALESCE (Phone, alt-phone, 'No Contact')
AS Contact-number
FROM Suppliers;

Supplier_id	Name	Contact-number
1	Global Goods	123456789
2	Best-Supplies	987654321
3	ValueSource	No Contact.

Table 15 User-Settings

Syntax

SELECT user_id, COALESCE (theme, 'light') AS theme-Set,
COALESCE (Language, 'English') AS Language-Set,
COALESCE (timezone, 'UTC') AS timezone-Set
FROM User-Settings;

User_id	theme-Set	Language-Set	timezone-Set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC.

Table 16: Maintenance.

Syntax

```
SELECT record-id, Machine-id, COALESCE (issue,  
'Unknown Issues') AS issue-log,  
COALESCE (technician, 'Not Assigned') AS  
technician-name  
FROM Maintenance.
```

Record-id	Machine-id	Issue-log	technician-name
1	M101	Overheating	Not Assigned
2	M102	unknown issue	Not Assigned.
3	M103	Jammed	Alex